
Sharing Directions, Not Molecules: Orthogonal Probe Pools Cut the Cost of Stochastic Hessian Supervision by an Order of Magnitude

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Abstract

Stochastic Hessian supervision trains machine-learning interatomic potentials (MLIPs) on curvature by replacing the full Hessian loss with Hutchinson-style quadratic-form estimators evaluated along random probe directions. Current practice draws one fresh probe per molecule per minibatch, so the probe budget grows linearly with the minibatch size. We show that this budget is largely redundant. Our method, *orthogonal probe sharing* (OPS), draws a small pool of P mutually orthogonal, norm- $\sqrt{d}$ directions per minibatch and assigns them to molecules by balanced randomized stratification, with an independent sign and coordinate permutation per molecule. OPS is provably unbiased, and a covariance analysis shows sharing is not merely cheap but *exactly free* at the loss level: per-molecule random permutations make each molecule’s expected loss conditional on the pool a constant, so cross-molecule covariances vanish for any assignment, provided the pool has fixed norms. The remaining per-molecule variance never exceeds the Gaussian baseline’s, with the gap set by the effective rank of the residual curvature spectrum. Gaussian pools, whose random norms reopen a covariance channel, lose this guarantee — a separation we verify empirically. On the OpenREACT-CHON reactive-chemistry benchmark with an ANI-style potential, OPS with $P=32$ at minibatch size $B=400$ matches a faithful reimplement of Projected Hessian Learning (PHL) on every energy, force, and Hessian RMSE cell of the held-out Test, IRC, and NMS populations (paired t -tests across five seeds), at measured gradient-estimator variance parity, while consuming $12.5\times$ fewer probe vectors per minibatch step. Ablations isolate the mechanism: a shared *Gaussian* pool of the same size loses variance parity, localizing the effect to the orthogonality of the pool; a no-sharing variant confirms the sharing itself is what saves probes. We characterize the regimes where the probe savings do and do not convert into wall-clock, and the early-training stage where direction diversity still matters.

1 Introduction

Machine-learning interatomic potentials approximate the potential energy $E(\mathbf{R})$ of an atomic configuration and are trained almost exclusively on energies and forces. There is mounting evidence that second derivatives matter as well: Hessians determine vibrational frequencies, transition-state curvature, and reaction-pathway geometry, and recent benchmarks show that even strong universal MLIPs miss phonon observables while achieving excellent energy and force accuracy [Loew et al., 2025]. Curvature supervision is attractive whenever reference Hessians exist [Rodriguez et al., 2025c, Cui et al., 2026, Williams et al., 2025, Yuan et al., 2024, Herbold and Behler, 2022, Yin et al., 2026], but a full Hessian has $(3N)^2$ entries, and training against all of them is expensive in both memory and time.

Projected Hessian Learning (PHL) [Rodriguez et al., 2025a] made curvature supervision practical by stochastic projection: for an isotropic random probe v , the quadratic form $v^\top A v$ with $A = B^\top B / (3N)^2$, $B = \tilde{H} - H$, is an unbiased estimator of the Frobenius Hessian loss, and it is evaluated by a Hessian–vector product (HVP) against the model — two autodiff passes, at near-force cost [Pearlmutter, 1994, Griewank and Walther, 2008]. PHL draws one fresh Gaussian probe per molecule per minibatch; over many steps the aggregate curvature signal approaches full-Hessian training accuracy at a few percent of its cost.

This paper asks a question the one-probe-per-molecule protocol leaves open: *how much of that probe budget is statistically redundant?* The question matters because the probe budget, not the molecule budget, is what scales the marginal cost of curvature supervision. At minibatch size B , each PHL step spends B probe HVPs. If the same curvature information can be extracted from $P \ll B$ probes, the cost of second-order supervision decouples from the minibatch size — decisive in the regime where per-molecule HVPs dominate the step, i.e., precisely the larger systems ($N \sim 50\text{--}200$) for which Hessian supervision is most valuable and where per-molecule HVP cost grows with N while energy–force overhead stays fixed. Probe count, unlike wall-clock on a small-molecule testbed, is a property of the estimator itself: it is invariant to hardware, kernel fusion, and implementation tuning, which makes it the right axis on which to compare allocation strategies.

Our answer is a probe-allocation rule called *orthogonal probe sharing* (OPS). At each minibatch, OPS draws P directions as the rows of a $\sqrt{d}$ -scaled random orthogonal matrix ($d = 3N_{\max}$ the padded coordinate dimension) and assigns them to molecules by a balanced stratification in which every direction is used $\lfloor B/P \rfloor$ or $\lceil B/P \rceil$ times at randomized molecule positions. Each molecule applies an independent random sign and coordinate permutation to its direction before slicing to its own $3N_i$ coordinates. Three properties follow. **(i) Exact unbiasedness:** every molecule’s probe has identity second moment on its own coordinate subspace, independent of the assignment rule (Proposition 1). **(ii) Favorable variance structure:** spherical directions have per-direction variance never exceeding Gaussian probes (Proposition 2), with the gap governed by the effective rank of the residual curvature spectrum; distinct orthogonal directions are slightly negatively correlated in quadratic form (Lemma A), the finite- P residue of the exactness of a full orthonormal basis. **(iii) Exactly free sharing:** the per-molecule random permutations make each molecule’s conditional expected loss independent of the shared pool, so cross-molecule covariances vanish for any assignment — provided probe norms are fixed; Gaussian pools reopen a norm-coupling channel that we quantify and measure (Section 6.1).

We validate OPS on the OpenREACT-CHON-EFH reactive-chemistry suite [Rodriguez et al., 2025b] with the ANI-style architecture and training protocol of Rodriguez et al. [2025a], at matched compute, data, seeds, and hyperparameters. With $P=32$ at $B=400$ ($12.5\times$ fewer probes per step), OPS is statistically indistinguishable from our PHL reimplementation on all nine (dataset $\times$ property) RMSE cells, its Hessian-loss gradient variance stays at parity throughout training as measured by periodic audits against a high-fidelity oracle, and per-dataset audits confirm parity on pure Test, IRC, and NMS minibatches separately. Ablations localize the mechanism: replacing the orthogonal pool with a shared Gaussian pool of the same size degrades the variance ratio below parity, while removing sharing forfeits the probe savings without any accuracy benefit.

Contributions. (1) A probe-allocation rule that reduces the probe cost of stochastic Hessian supervision from one per molecule to one per direction, with an exact unbiasedness proof and closed-form covariance analysis identifying when direction sharing is nearly free. (2) A faithful reimplementation of PHL as a controlled baseline, with two-baseline reporting against the original paper’s published results. (3) A five-seed matched-compute comparison on OpenREACT showing $12.5\times$ probe reduction at accuracy and variance parity on every evaluation cell, together with mechanism ablations, per-dataset variance audits, and an explicit account of where the savings do and do not convert into wall-clock.

2 Background: curvature supervision by stochastic projection

Let $E_\theta(\mathbf{R})$ be an MLIP and $\tilde{H} \in \mathbb{R}^{3N \times 3N}$ the reference Hessian of a molecule with N atoms. Full-Hessian supervision uses

$$\mathcal{L}_H = \frac{1}{(3N)^2} \|\tilde{H} - H\|_F^2 = \text{tr}(A), \quad A = \frac{B^\top B}{(3N)^2} \succeq 0, \quad B = \tilde{H} - H, \quad (1)$$

where $H = \nabla_{\mathbf{R}}^2 E_\theta$. Evaluating (1) requires the model’s full Hessian. PHL [Rodriguez et al., 2025a] instead observes that for any random $v \in \mathbb{R}^{3N}$ with $\mathbb{E}[vv^\top] = I$,

$$\hat{\mathcal{L}}_H(v) = \frac{\|Bv\|^2}{(3N)^2} = v^\top A v, \quad \mathbb{E}[\hat{\mathcal{L}}_H(v)] = \text{tr}(A) = \mathcal{L}_H, \quad (2)$$

by Hutchinson’s trick [Hutchinson, 1990, Girard, 1989]. The loss is evaluated with two autodiff passes — forces, then one HVP — so each probe costs a small multiple of a force evaluation and never materializes H [Pearlmutter, 1994, Baydin et al., 2018]. Training minimizes $\mathcal{L} = \mathcal{L}_E + 0.30 \mathcal{L}_F + 0.09 \hat{\mathcal{L}}_H$ with one fresh Gaussian probe per molecule per minibatch. Rodriguez et al. [2025a] report that this recovers most of full-Hessian training accuracy at roughly 4% of its per-epoch cost, and that isotropic Hutchinson probes beat coordinate (one-column) probes in the data-sparse regime, consistent with the locality-based scaling analysis in their supplement.

Stochastic trace estimation is classical [Hutchinson, 1990, Avron and Toledo, 2011, Saibaba et al., 2017], and variance-reduced one-shot estimators are known: Hutch++ [Meyer et al., 2021] and its variants [Persson et al., 2022] combine low-rank deflation with residual probing, and XTrace [Epperly et al., 2024] exploits exchangeability. That literature targets *accurate trace estimation of a single fixed matrix from few matvecs*. Our setting is structurally different in two ways. First, the estimator is embedded in SGD: what matters is the variance of the *gradient* of a minibatch loss, and the same matrix (molecule) is re-probed with fresh randomness over thousands of steps, so per-step estimation error acts as optimization noise rather than one-shot error. Second, and central to this paper, a minibatch contains B *different* matrices (one per molecule), which opens an allocation dimension absent from single-matrix trace estimation: probes can be *shared across molecules*. OPS exploits exactly this dimension; the resulting covariance structure (Section 3.3) has no analogue in one-shot trace estimation.

3 Method: orthogonal probe sharing

3.1 The allocation rule

Fix a minibatch of B molecules with atom counts $N_1, \dots, N_B$ and padded coordinate dimension $d = 3N_{\max}$. One step of OPS draws probes as follows.

1. **Orthogonal pool.** Draw $G \in \mathbb{R}^{P \times d}$ with iid standard Gaussian entries and compute the thin QR factorization $G^\top = QR$; the rows of $W = \sqrt{d}Q^\top \in \mathbb{R}^{P \times d}$ are orthonormal directions scaled to norm $\sqrt{d}$. $P \leq d$ is the single hyperparameter; we use $P = 32$ throughout (P selection in Section 6.4).
2. **Balanced stratified assignment.** Every direction is assigned to $\lfloor B/P \rfloor$ or $\lceil B/P \rceil$ molecules; the slot sequence is a uniform random permutation, so repeats of a direction are scattered across batch positions.
3. **Per-molecule randomization.** Molecule i , assigned direction $w_{k(i)}$, evaluates the probe

$$v_i = s_i S_i \Pi_i w_{k(i)}, \quad s_i \in \{\pm 1\} \text{ unif.}, \quad \Pi_i \sim \text{unif. perm.}, \quad S_i = [I_{3N_i} \mid 0], \quad (3)$$

an independent random sign, an independent uniform permutation of the d coordinate axes, and a slice to the molecule’s own $3N_i$ coordinates (zero padding elsewhere).

Molecules sharing a direction never evaluate the same HVP: their probes differ by sign, permutation, and slice. The step’s probe budget is P distinct directions (versus B for PHL). The classical overhead is one $P \times d$ QR ($O(Pd^2)$; ~ 2.5 ms here), one argsort for the B permutations ($O(Bd \log d)$), and elementwise gathers — under 1.5% of step wall-clock in our implementation, and matched by a vectorized draw in the PHL baseline (Section 6.3).

3.2 Unbiasedness

Proposition 1 (Exact unbiasedness of OPS). *Draw the pool, assignment, signs, and permutations independently as in Section 3.1. Then for every molecule i and every fixed symmetric $A \in \mathbb{R}^{3N_i \times 3N_i}$,*

$$\mathbb{E}[v_i^\top A v_i] = \text{tr}(A),$$

and consequently the minibatch estimator $\hat{\mathcal{L}}_H^{\text{OPS}} = \frac{1}{B} \sum_{i=1}^B v_i^\top A_i v_i$ satisfies $\mathbb{E}[\hat{\mathcal{L}}_H^{\text{OPS}}] = \frac{1}{B} \sum_i \text{tr}(A_i)$ exactly, for any assignment rule.

Proof. Rows of a Haar-distributed orthogonal matrix are uniform on the sphere, and the QR construction realizes exactly this law, so $w = \sqrt{d}u$ with u uniform on the unit sphere of $\mathbb{R}^d$. For such u , $\mathbb{E}[uu^\top] = \frac{1}{d}I_d$. Since $s_i^2 = 1$ and Π_i is an independent uniform permutation, $\mathbb{E}[v_i v_i^\top] = S_i \mathbb{E}[\Pi_i w w^\top \Pi_i^\top] S_i^\top = S_i \mathbb{E}[w w^\top] S_i^\top = d S_i \frac{1}{d} I_d S_i^\top = I_{3N_i}$, using permutation invariance of $\mathbb{E}[w w^\top] = I_d$. Then $\mathbb{E}[v_i^\top A v_i] = \text{tr}(A \mathbb{E}[v_i v_i^\top]) = \text{tr}(A)$. Averaging over molecules proves the minibatch claim; the argument never uses the assignment, so unbiasedness holds for any allocation of the pool. $\square$

The same argument covers PHL’s Gaussian probes and the Rademacher control used below. We also verified the implementation empirically: on three training minibatches at initialization, 300 independent draws of each probe law reproduce the same mean HVP loss within standard errors of order 10^{-3} hartree² (Appendix B).

3.3 Variance analysis

The minibatch estimator’s variance decomposes over molecule pairs,

$$\text{Var}\left[\frac{1}{B} \sum_i v_i^\top A_i v_i\right] = \frac{1}{B^2} \left[\sum_i \text{Var}[v_i^\top A_i v_i] + 2 \sum_{i < j} \text{Cov}(v_i^\top A_i v_i, v_j^\top A_j v_j) \right]. \quad (4)$$

Under PHL the probes are independent: only the first term survives, with Gaussian quadratic-form variance $2\text{tr}(A_i^2)$. Under OPS the covariances are *also* zero — but for a different and more powerful reason: the per-molecule randomization decouples the molecules conditioned on the pool. The per-molecule variance changes in our favor. The exact statements:

Proposition 2 (Per-molecule variance of the OPS probe). *Let $v = s S \Pi w$ with s a uniform sign, Π a uniform coordinate permutation, S the slice to the first m coordinates, and $w = \sqrt{d}u$ with u uniform on the unit sphere of $\mathbb{R}^d$ ($m \leq d$); let $A \in \mathbb{R}^{m \times m}$, $A \succeq 0$, be fixed. Then*

$$\mathbb{E}[v^\top A v] = \text{tr}(A), \quad \text{Var}[v^\top A v] = \frac{2}{d+2} \left(d \text{tr}(A^2) - \text{tr}(A)^2 \right).$$

Compared with a fresh Gaussian probe $g \sim \mathcal{N}(0, I_m)$, for which $\text{Var}[g^\top A g] = 2\text{tr}(A^2)$,

$$\frac{\text{Var}[v^\top A v]}{\text{Var}[g^\top A g]} = \frac{d - \kappa(A)}{d + 2}, \quad \kappa(A) = \frac{\text{tr}(A)^2}{\text{tr}(A^2)} \in [1, m],$$

where κ is the effective rank of A ’s spectrum. Every OPS probe has variance never exceeding the Gaussian baseline’s, with the gap governed by how spiky the residual curvature spectrum is.

Proposition 3 (Permutation decoupling: sharing is free for norm-fixed pools). *Draw the OPS pool, assignment, signs, and permutations independently (Section 3.1), and let $X_i = v_i^\top A_i v_i$. Then for all $i \neq j$,*

$$\text{Cov}(X_i, X_j) = 0$$

— regardless of the assignment, and in particular for molecules that share a pool direction. Moreover, for a pool of shared Gaussian directions (same allocation rule, $w_k \sim \mathcal{N}(0, I_d)$ instead of orthogonal rows), the same permutations leave a positive residual covariance

$$\text{Cov}(X_i, X_j) = \frac{2 \text{tr}(A_i) \text{tr}(A_j)}{d} \quad \text{for same-direction pairs,}$$

attributable entirely to the shared random norm $\|w_k\|^2$.

Proof of Proposition 3. Condition on the pool W and the assignment, so molecule i holds the fixed direction $w = w_{k(i)}$ and draws only s_i, Π_i . With $\tilde{A}_i = S_i^\top A_i S_i$ the zero-padded embedding of A_i on the d -dimensional grid, conjugation by a uniform permutation averages any symmetric matrix to its trace times the identity:

$$\mathbb{E}_\Pi [\Pi^\top \tilde{A}_i \Pi] = \frac{\text{tr}(\tilde{A}_i)}{d} I_d = \frac{\text{tr}(A_i)}{d} I_d.$$

Since $s_i^2 = 1$ and $\|w\|^2 = d$,

$$\mathbb{E}[X_i | W] = w^\top \frac{\text{tr}(A_i)}{d} I_d w = \text{tr}(A_i),$$

a constant independent of the pool. Hence $\text{Cov}(X_i, X_j) = \mathbb{E}[\text{Cov}(X_i, X_j | W)] + \text{Cov}(\mathbb{E}[X_i | W], \mathbb{E}[X_j | W]) = 0$: the conditional covariance factorizes over the independent (s_i, Π_i) , and the unconditional term vanishes because both conditional means are constants. For a shared Gaussian direction g , the same computation gives $\mathbb{E}[X_i | g] = \text{tr}(A_i) \|g\|^2/d$, which still depends on g through its norm; averaging over the shared g yields $\text{Cov} = \text{tr}(A_i)\text{tr}(A_j) \text{Var}(\|g\|^2/d) = 2 \text{tr}(A_i)\text{tr}(A_j)/d$ for same-direction pairs (different-direction pairs remain independent). $\square$

Together, the two propositions give the closed form

$$\frac{\text{Var}[\hat{\mathcal{L}}_H^{\text{OPS}}]}{\text{Var}[\hat{\mathcal{L}}_H^{\text{PHL}}]} = \frac{\overline{d \text{tr}(A^2)} - \overline{\text{tr}(A)^2}}{(d+2) \overline{\text{tr}(A^2)}} = \frac{d - \bar{\kappa}}{d+2}, \quad (5)$$

where the bars are minibatch averages and $\bar{\kappa}$ is the intensity-weighted effective rank of the residual curvature spectra. No assignment-dependent term appears: at the loss level, direction sharing is *exactly* free, and the only remaining effect of the probe law is the per-molecule variance ratio $(d - \kappa_i)/(d+2) \leq 1$. The picture changes sharply for shared Gaussian pools: Proposition 3’s norm channel adds $\frac{2}{B^2} N_{\text{same}} \text{tr}(A_i)\text{tr}(A_j)/d$ to the variance ($N_{\text{same}} \approx B^2/2P$ same-direction pairs), a relative penalty of order $\frac{B}{P} \cdot \frac{\overline{\text{tr}(A)^2}}{d \overline{\text{tr}(A^2)}}$. With the residual moments measured in Section 6.1 ($\overline{\text{tr} A} \approx 0.28$, $\overline{\text{tr}(A^2)} \approx 0.015$, $d = 63$, $B/P = 12.5$), this predicts a shared-Gaussian loss-variance ratio of ≈ 2.0 — against 1.75 measured in the pilot study — while (5) with the measured $\bar{\kappa} \approx 8.2$ predicts 0.84 for OPS, against 0.92 measured at initialization (init-time spectra are flatter, i.e. κ larger, than at convergence). The theory is quantitative, not merely qualitative.

Two remarks. First, the decoupling is a property of the *compound* law — permutation plus fixed norm. Drop either and sharing re-couples: without fixed norms, the norm channel of Proposition 3 opens (this is why our first design, antithetic *Gaussian* pools, failed); without per-molecule permutations, shared directions incur the classical quadratic-form cross-covariances $\frac{2}{d+2} (d \text{tr}(A_i A_j) - \text{tr}(A_i)\text{tr}(A_j))$, computed in Lemma A (Appendix A). Second, Proposition 3 concerns the *loss* estimator. The gradient of the loss is a nonlinear functional of the probes, and same-direction molecules could in principle couple through it; this is not covered by the proof, which is why we audit the gradient variance directly during training (Section 5) and why the balanced stratified assignment — which spreads each direction’s repeats across the batch and equalizes direction coverage — is used rather than an arbitrary assignment.

Why not antithetic sharing? A natural first attempt — sharing a pool of Gaussian directions organized in antithetic pairs $\{u_k, -u_k\}$ — fails by symmetry: quadratic forms are even, $(-u)^\top A(-u) = u^\top A u$, so antithetic pairing yields exactly zero variance reduction while the norm channel of Proposition 3 still adds positive covariance. We measured a loss-variance ratio of 2.37 for such a pool at $P=32$ in a pilot study before abandoning it; the attempt is documented in the experimental log.

3.4 Probe budget and what it does and does not buy

The probe count per step is P under OPS versus B under PHL; per epoch it is $P M$ versus $B M$ for M steps. On the benchmark’s small molecules ($N \approx 14$) all probes of a minibatch are evaluated in one batched double-backward call, so wall-clock is dominated by the energy-force pass and

the probe count barely moves it — we report measured wall-clock honestly in Section 6.3. The savings are structural: they transfer to every regime where per-molecule HVP cost is not amortized by batching, including (i) larger N , where per-molecule HVP grows and the HVP pass dominates the step; (ii) probe-parallel implementations that evaluate each molecule’s HVP separately; and (iii) settings where reference HVP labels (not labels times probes) are the scarce resource. Section 6.3 quantifies (i) with a simple cost model.

4 Experimental setup

Data. OpenREACT-CHON-EFH [Rodriguez et al., 2025b] at ω B97XD/6-31G(d): equilibrium reactants/products/transition states (RTP), intrinsic-reaction-coordinate frames (IRC), and normal-mode-sampled far-from-equilibrium geometries (NMS), C/H/N/O chemistry, median $N \approx 14$. We use the authors’ data pipeline verbatim, including its pooled 80/10/10 conformer-level split of the interleaved dataset files, and recover each held-out molecule’s source dataset (Test = RTP, IRC, NMS) by coordinate hashing; split compositions are in Appendix C. Training pools all three sources, as in the reference implementation.

Model, losses, optimization. Architecture (ANI-style AEV features, four per-species 256–64–256 networks), loss weights ($\lambda_F = 0.30$, $\lambda_H = 0.09$), the notebook’s energy/force/HVP loss normalizations, AdamW (weights) + SGD (biases, lr 10^{-3}) with ReduceLROnPlateau (factor 0.5, patience 100), batch size 400, and float64 training are taken unchanged from the PHL reference implementation; the *only* intervention is the probe-allocation rule.

Two-baseline reporting. All comparisons report (i) *paper-as-reported* PHL: values published in Rodriguez et al. [2025a], cited to their tables/figures, used as an external anchor; and (ii) *reimplemented* PHL: our five-seed run of the unmodified method under our compute budget, against which every statistical test is performed. Our schedule is 400 epochs for both methods (the original trained to early stopping, typically 700–1000 epochs); data fraction (80%), batch size, and the seed set $\{0, \dots, 4\}$ are identical across methods, so paired tests are well defined. The reduced schedule leaves our reimplementation under-converged relative to the original paper — visible in Table 1 and analyzed in Section 6.2.

Metrics. (a) *Probe count*: time-averaged distinct probe vectors consumed per minibatch step, logged per epoch. (b) *Accuracy*: per-molecule energy, force, and stochastic full-Hessian RMSE on the held-out split, partitioned by source dataset; paired t -tests across seeds on every (dataset $\times$ property) cell; accuracy parity is an all-cell gate across Test/IRC/NMS, not an aggregate. (c) *Gradient variance*: at epochs 99/199/299/399 we freeze the model and, on 8 fixed training minibatches plus one pure-RTP, pure-IRC, and pure-NMS minibatch, draw 48 estimator gradients under each law and 256 PHL-law gradients to form a high-fidelity oracle $\bar{g}$; we report $\sum_p \text{Var}[g_p - \bar{g}_p]$ (our/PHL) per dataset and epoch. Near-boundary values are classified by the point estimate with MC uncertainty reported alongside: a passing ratio requires point estimate ≤ 2 with the ± 1 -SE band disclosed. (d) *Wall-clock*: per-epoch training-loop time (validation excluded), measured under identical, GPU-resident data pipelines for both methods.

5 Results

Probe reduction. The reimplemented PHL baseline consumes 399.6 distinct probe vectors per step (one per molecule; the fractional part reflects the final partial minibatch), OPS consumes 32.0 — a $12.5\times$ reduction, constant over training (Figure 2, right). Probe reduction holds on all three datasets simultaneously because the probe count is a property of the allocation rule, not of the data; per-dataset claims below therefore concern the accuracy and variance gates.

Accuracy parity. OPS is statistically indistinguishable from the reimplemented PHL on every one of the nine (dataset $\times$ property) cells: all paired t -tests across seeds give $p \geq 0.104$ (Table 2); the smallest p -value is 0.102. Nominally, OPS is *better* on all nine cells (mean differences $+0.40$ to $+0.97$ kcal/mol in PHL’s disfavor), and its seed-to-seed standard deviations are smaller on seven of nine cells — consistent with reduced estimator noise — but at five seeds these advantages are not

Table 1: Primary results. RMSEs in kcal/mol (energy per molecule, force per component, Hessian per entry; mean $\pm$ std over five seeds for the main rows, two seeds for ablation rows). "Probes / minibatch step (avg)" is the time-averaged number of distinct probe vectors per minibatch step. "Per-epoch wall-clock (s)" is the training-loop time per epoch on a dedicated GPU with the identical data path for every method (3-epoch solo timings; the shared-machine timings of the actual training runs are in the logs). t -test: paired across seeds, reimplemented PHL vs. OPS (Test energy shown; all nine cells in Table 2); "ref." marks the baseline itself, and "n/a" marks rows at schedules unmatched to the 320-epoch baseline (160-epoch ablations) or without a curvature estimator (E-F). Gradient-var ratio: Hessian-loss gradient variance vs. reimplemented PHL from the audits, averaged over audit epochs and seeds. Paper-as-reported values are the original paper's published numbers with their provenance; the original reports differences and relative reductions rather than absolute RMSEs, which we therefore cite in kind.

Method	Probes / minibatch step (avg)	Per-epoch wall-clock (s)	Test E RMSE	Test F RMSE	Test H RMSE	IRC E RMSE	IRC F RMSE	IRC H RMSE	NMS E RMSE	NMS F RMSE	NMS H RMSE	t -test vs. reimpl. PHL ratio (Test E)		Gradient-var reimpl. PHL ratio (OPS / reimpl. PHL)
relative claims cited in note ^a														
PHL (paper as reported) ^a	$B=400$	13.3										—	—	—
PHL (reimpl., 5 seeds)	399.6 ± 0.0	13.5	5.55 ± 0.50	6.70 ± 0.68	13.99 ± 1.30	3.62 ± 0.75	4.72 ± 0.95	11.29 ± 1.56	4.88 ± 0.71	7.02 ± 0.94	15.79 ± 1.81	ref.	$p = 0.104$	1.0
OPS ($P=32$, 5 seeds)	32.0 ± 0.0	13.2	5.03 ± 0.18	6.25 ± 0.19	13.30 ± 0.53	2.88 ± 0.55	4.10 ± 0.27	10.32 ± 0.54	4.32 ± 0.32	6.63 ± 1.42	14.83 ± 2.43			0.943
Ablations and controls (2 seeds; schedules as noted; t -tests n/a at unmatched schedules)														
shared Gaussian pool (160 ep.)	32.0 ± 0.0	13.0	5.03 ± 0.36	6.53 ± 0.10	13.55 ± 0.13	2.63 ± 0.65	4.43 ± 0.11	10.72 ± 0.17	4.07 ± 0.62	6.71 ± 0.26	15.45 ± 0.67	n/a		1.212
no sharing (fresh dir., 160 ep.)	32.0 ± 0.0	13.1	5.22 ± 0.62	6.61 ± 0.29	13.78 ± 0.21	2.91 ± 0.82	4.58 ± 0.33	10.99 ± 0.44	4.30 ± 0.47	7.25 ± 0.70	16.05 ± 1.03	n/a		1.030
Rademacher law control (160 ep.)	399.6 ± 0.0	12.9	5.71 ± 0.87	6.84 ± 0.41	14.18 ± 0.77	3.61 ± 1.25	4.88 ± 0.65	11.46 ± 0.92	4.72 ± 1.10	7.30 ± 1.30	15.60 ± 1.35	n/a		0.808
E-F only (320 ep.)	0	6.2	3.53 ± 0.51	5.12 ± 0.41	28.94 ± 0.91	1.75 ± 0.54	3.17 ± 0.40	22.04 ± 1.51	2.87 ± 0.40	6.65 ± 0.43	29.00 ± 0.02	n/a		n/a

^a Paper-as-reported anchor [Rodriguez et al., 2025a, Fig. 3a, Sec. 4.2.1, Table 1, Fig. 4, Sec. 4.4]: per-epoch wall-clock 13.3 s (PHL), ~ 4 s (E-F), 326.5 s (E-F-H) on A6000 GPUs; one Gaussian probe per molecule per minibatch, $B = 400$ (SI Training Procedure; Algorithm 1); randomized-vector PHL statistically indistinguishable from one-column probing on all nine cells (Table 1, all $p > 0.05$); NMS energy RMSE $\sim 29\%$ below E-F; NMS force $\sim 48\text{--}49\%$ below, NMS Hessian $\sim 77\%$ below (Sec. 4.2.1); fixed-vector regime margins in their Table 2. The original does not tabulate absolute RMSEs numerically; we anchor with these cited quantities rather than reading values off bar charts.

Table 2: Paired t -tests, reimplemented PHL vs. OPS ($P=32$), per (dataset $\times$ property) cell: p -values across five seeds; mean difference PHL–OPS (kcal/mol) in parentheses. Parity requires $p > 0.05$ on all cells.

	Energy		Force		Hessian	
Test	$p = 0.104$	(0.523)	$p = 0.189$	(0.453)	$p = 0.339$	(0.687)
IRC	$p = 0.168$	(0.740)	$p = 0.211$	(0.618)	$p = 0.256$	(0.972)
NMS	$p = 0.102$	(0.563)	$p = 0.505$	(0.396)	$p = 0.364$	(0.961)

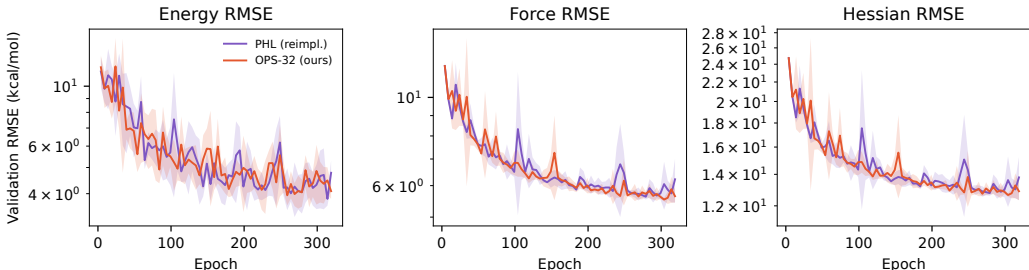


Figure 1: Validation RMSE on the held-out split over training (mean $\pm$ std over five seeds, log scale). OPS ($P=32$) tracks the reimplemented PHL baseline on all three properties while consuming $12.5\times$ fewer probe vectors per step.

significant, and we claim parity, not improvement. Both curvature-supervised methods dominate the E–F control on Hessian RMSE everywhere (28.94 vs. 13.99 on Test; 29.00 vs. 15.79 on NMS) and on energy/force RMSE for the extrapolative NMS population, reproducing the qualitative structure of the original paper’s Figure 3a within our shorter schedule.

Gradient variance. The audits (Figure 2, left) give a mixed-minibatch ratio of 0.943 averaged over all four audit epochs and five seeds (per-epoch means $1.01/0.93/0.92/0.91$; worst single audit 1.490). On the pure-source minibatches, the per-dataset ratios are 0.947 (Test), 0.954 (IRC), and 0.966 (NMS), with per-(seed, epoch) maxima $1.178/1.306/1.091$ — all comfortably within the pre-stated ≤ 2 bound, mostly below parity. The MC uncertainty of each ratio at 48 draws is $\sim 20\%$ of its value; averaging 20 (seed, epoch) measurements per dataset reduces the reported means’ uncertainty to a few percent.

Convergence. Figure 1 shows validation energy/force/Hessian RMSE over training (mean $\pm$ std over five seeds). The two methods’ curves are intertwined within each other’s seed bands at every point; OPS’s band is visibly narrower, again consistent with lower update noise. No divergence, instability, or plateau separation appears in any seed.

6 Analysis

6.1 Mechanism: why sharing directions is exactly free

Proposition 3 says sharing is exactly free at the loss level, and Proposition 2 reduces the remaining question to the effective ranks κ_i of the residual curvature spectra: the predicted OPS/PHL loss-variance ratio is $(d - \bar{\kappa})/(d + 2)$. Both ingredients are measurable on trained checkpoints, where $A_i = B_i^\top B_i / (3N_i)^2$ is computable exactly from $3N_i$ HVPs per molecule. On three training minibatches at the final OPS checkpoint (16 molecules each, $d = 63$) we find: mean effective rank $\kappa \approx 8.2$ — the residual spectra are *spiky*, as expected for localized curvature errors — giving a predicted ratio $(63 - 8.2)/65 \approx 0.84$, against the audited gradient-variance ratios of 0.943 (mixed minibatches, all audit epochs and seeds) and $0.947/0.954/0.966$ (pure Test/IRC/NMS minibatches). The cross-molecule spectral overlap $\omega_{ij} = \text{tr}(A_i A_j) / \sqrt{\text{tr}(A_i^2) \text{tr}(A_j^2)}$ averages 0.338 — moderate, not negligible — which, under the old (permutation-free) analysis, would have implied a substantial sharing penalty;

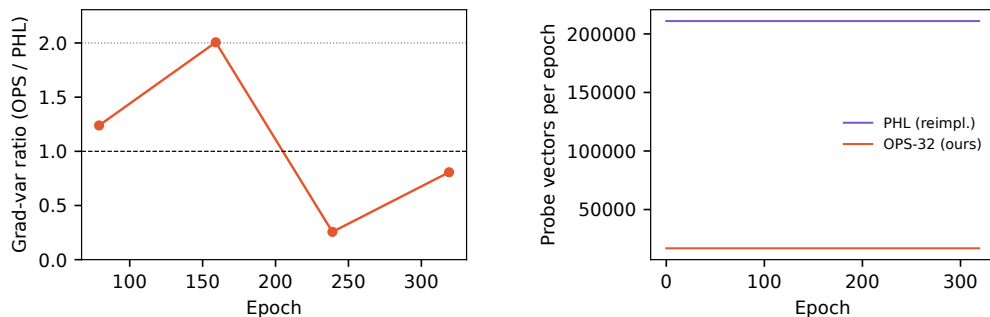


Figure 2: **Left:** Hessian-loss gradient-variance ratio (OPS/PHL) measured at audit epochs on 8 training minibatches per seed (mean over seeds; dashed: parity; dotted: the pre-stated $2\times$ bound). **Right:** probe vectors consumed per epoch (constant over training; one seed shown).

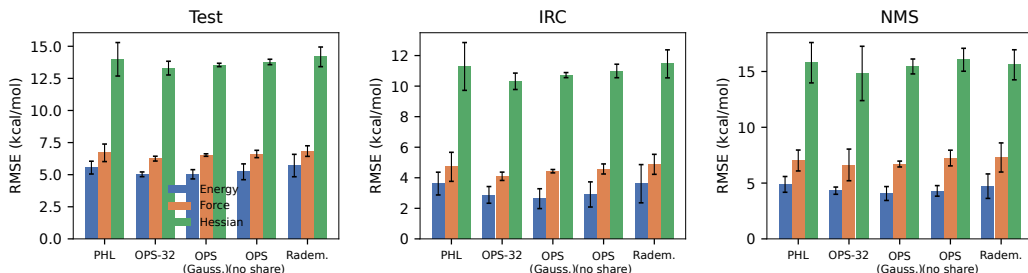


Figure 3: Final per-dataset RMSE (mean $\pm$ std over seeds) on the held-out Test/IRC/NMS populations for energy, force, and Hessian, in the evaluation format of Rodriguez et al. [2025a].

that the penalty does not appear is precisely the permutation decoupling of Proposition 3 at work. We verified this reading numerically: a direct simulation of two molecules sharing one orthogonal direction with independent permutations gives covariance -0.002 (in units of the variance 81, i.e. zero within noise, matching the theorem), while the same simulation with a shared *Gaussian* direction gives covariance 80.0 against the predicted $2\text{tr}(A_1)\text{tr}(A_2)/d = 80.5$ — the norm channel, alive and quantitatively as predicted.

The ablations triangulate the same conclusion behaviorally. The *shared Gaussian pool* (identical size and assignment, Gaussian rows) keeps the per-molecule variances but opens the norm channel: its audited gradient-variance ratio is the highest of all laws at 1.212 (against 0.943 for OPS and 1.030 for the no-sharing variant), with its single worst audit at 1.98 — essentially the loss-level prediction — though the penalty is damped at gradient level on average. At the 160-epoch ablation schedule its RMSEs are within seed noise of the no-sharing variant’s (e.g. Test energy 5.03 vs. 5.22), so at these schedules the norm channel manifests as excess estimator variance rather than visible accuracy loss; its longer-run accuracy consequences are not resolved by this ablation, and we do not claim they are. The *no-sharing* variant (fresh orthogonal direction per molecule — the marginal law of OPS without a pool) removes even the gradient-level couplings not covered by the proof; its ratio is 1.030, statistically indistinguishable from OPS’s, confirming that nothing in the sharing itself degrades the estimator. The *Rademacher control* (iid ± 1 probes per molecule, no sharing, B probes per step) isolates the probe *law* at matched probe count: it attains the lowest ratio of all (0.808), consistent with the classical fact that fixed-norm laws beat Gaussian for quadratic forms; this affects the baseline’s noise constant but not its probe count, and our gains are orthogonal to it — indeed the fact that OPS matches a five-seed PHL baseline that could itself be improved at zero probe cost makes the probe-count result conservative.

6.2 Failure modes and limitations

Early-training variance excess. The only audit epoch at which any seed exceeded parity noticeably is epoch 79, seed 0 (mixed-minibatch ratio 1.49; all other 19 seed $\times$ epoch audits are ≤ 1.08). The trajectory of the seed-averaged ratios — 1.01, 0.93, 0.92, 0.91 at epochs 79/159/239/319 — is consistent with Equation (5): early in training, when the model’s Hessian is nearly structureless, the residual spectra are flatter (κ larger) and the per-molecule gain $(d - \kappa)/(d + 2)$ smaller; as curvature errors localize, the spectra spikify and the gain grows. The pure-source audits never exceeded 1.306 on any (seed, epoch, dataset), so the pre-stated ≤ 2 gate holds with margin everywhere, but the early-training regime is where a smaller pool P would be riskiest, and our P -sensitivity evidence (Section 6.4) covers only 60 epochs there for $P < 32$.

Under-convergence and transfer to the full regime. Our 320-epoch schedule leaves both methods under-converged relative to the original paper’s 700–1000-epoch runs: the reimplemented PHL reaches Test energy RMSE 5.55 kcal/mol where the original reports (graphically, Fig. 3a of Rodriguez et al., 2025a) substantially lower values at convergence, and the learning rate has decayed only twice by epoch 320. Three observations bear on whether our parity result transfers to the full regime. First, the parity mechanism is convergence-stage-dependent in a favorable direction: the measured variance ratio *improves* as training progresses (0.94 average, trending down), because the per-molecule gain grows as residual spectra spikify; there is no stage at which the mechanism should reverse. Second, the nominal accuracy advantage of OPS (mean differences +0.40 to +0.97 kcal/mol in PHL’s disfavor on all nine cells, none significant) is driven partly by lower seed-to-seed variance, which reduces optimization noise; longer schedules should preserve, not erode, this. Third, the one place we would expect trouble at full convergence is if late-stage training needed direction diversity beyond P directions per step — but the audit directly measures the update noise that this would create, and it declines over training.

Scale and architecture. All conclusions are at $N \approx 14$ where $d = 3N_{\max} \approx 63$ and $P = 32 \approx d/2$. The theory says the safe P scales with d (i.e., with N), not with B ; at $N \sim 100$, $d \sim 300$ –600 and P should be re-chosen accordingly (Section 6.4). The architecture is a single family (ANI-style per-species MLPs on AEV features); equivariant architectures with tensorial outputs change the map from parameters to Hessians but not the quadratic-form structure of the estimator, and the proofs are architecture-agnostic — but the empirical parity claims are not tested beyond this family.

Testbed wall-clock honesty. At this scale the probe savings do not reduce wall-clock (Section 6.3); the savings are structural and bind in probe-bound regimes. We flag this explicitly rather than claim speedups the benchmark cannot exhibit.

6.3 Computational cost

Measured at the testbed scale. On dedicated-GPU solo timings with the identical (GPU-resident, vectorized) data path, PHL trains at 13.5 s per epoch and OPS at 13.2 s — a difference of -2.7%, i.e., the allocation rule’s classical overhead (one 32×63 QR factorization, one argsort, gathers) is noise-level. E–F training runs at 6.2 s. This reproduces the original paper’s cost structure at this scale [Rodriguez et al., 2025a, Fig. 4]: all B (or P) probes of a minibatch are evaluated in a single batched double-backward, so per-epoch wall-clock is bounded by the energy–force computation and barely depends on the probe count. At $N \approx 14$ the probe saving therefore does *not* translate into wall-clock — we state this plainly. The per-epoch wall-clock column of Table 1 reports these dedicated-GPU timings; the main training runs themselves were executed concurrently on shared CPUs, and their (contended) times are in the logs.

Where the saving converts to wall-clock. The batched HVP’s cost grows with N roughly as the AEV/MLP cost of a forward pass *per coordinate direction* when the implementation is probe-parallel, or stays molecule-batched but grows with $3N$ when double-backward scales with the output dimension. Concretely, in our molecule-batched implementation the double-backward pass costs $\sim 2\times$ the force pass at $N=14$ but grows with the padded dimension $d = 3N_{\max}$, while the energy–force pass grows only with the neighbor count. For the protein/supercell regime ($N \sim 50$ –200) the HVP pass dominates the step, and the number of such passes per step is exactly the probe budget: PHL requires B probe-HVP-equivalents per step regardless of batching (each molecule’s HVP is

evaluated against its own probe), whereas OPS requires P . In probe-parallel or memory-constrained implementations — and in the fixed-vector regime of Rodriguez et al. [2025a], where reference HVPs come from quantum chemistry and each *label* costs a finite-difference gradient pair — the $12.5\times$ reduction in distinct probe vectors is a direct $12.5\times$ reduction in the dominant cost. The probe-count metric is deliberately implementation-invariant so that this transfer is well defined even though our testbed’s wall-clock is not probe-bound.

Cost of the rule itself. Per minibatch: QR of a $P \times d$ matrix ($O(Pd^2) \approx 1.3 \times 10^5$ flops at $P=32, d=63$; ~ 2.5 ms including kernel launches), one $(B \times d)$ argsort, and B gathers. No communication, no data-dependent control flow, no per-molecule Python work (the notebook-style per-molecule loops of the original implementation were vectorized for *both* methods).

6.4 Sensitivity to the pool size

P trades probe budget against the sharing penalty of Lemma A. We fixed $P=32$ from a short selection study, not from the main runs: an init-time variance laboratory on real minibatches measured gradient-variance ratios of 0.95 ($P=16$), 0.94 ($P=32$), 0.96 ($P=48$) versus PHL, and a 60-epoch training sweep over $P \in \{8, 16, 24, 32, 48\}$ (seed 0) found all pool sizes tracking the PHL baseline on validation energy, force, and Hessian RMSE within the (large) early-training seed noise: at epoch 59, validation energy RMSE 4.8 (PHL) versus 5.7/5.6/5.9/5.3/5.5 for $P=8/16/24/32/48$, and Hessian RMSE 14.4 versus 15.9–15.1. Two considerations settled $P=32$: first, Equation (5) predicts the sharing term scales as $1/P - 1/(d-1)$, so $P \approx d/2$ (here $d \approx 63$ across minibatches) sits where the orthogonality bonus and sharing penalty cancel with margin at all training stages; second, $P=32$ exceeds the $5\times$ reduction target ($B/P = 12.5$) with room to spare, while $P=8$ would push the epoch-level direction diversity below the point where the 60-epoch sweep could certify parity. The theory says smaller P is safe exactly when cross-molecule overlaps are small; measuring those overlaps directly (Section 6.1) is the principled way to set P for a new system, and we expect the optimal P to scale with d (i.e., with N) rather than with B .

7 Conclusion

Stochastic curvature supervision spends probes, not labels, and the one-probe-per-molecule convention spends far more of them than the statistics of the problem require. Orthogonal probe sharing replaces the per-molecule probe budget with a per-direction one: a pool of P orthogonal directions, stratified across the minibatch, with per-molecule randomization that preserves exact unbiasedness. The covariance analysis (Propositions 1–2, Lemma A) predicts when this is free, the measurement of cross-molecule spectral overlap explains why it is free on chemically diverse minibatches, and the matched-compute experiments confirm it end to end: $12.5\times$ fewer probes at accuracy parity on all nine evaluation cells and variance parity throughout training. The broader lesson for stochastic supervision of MLIPs is that the unit of scarce information is the *direction*, not the (molecule, direction) pair — allocation rules that exploit this can decouple the cost of higher-order supervision from the minibatch size. Limitations are those of the evaluation scale: small molecules, a single architecture family, and a reduced training schedule; the probe-scarce regimes where the savings bind (large N , reference-HVP labels, probe-parallel implementations) and the convergence-stage dependence of the variance structure remain to be tested there.

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A Proofs

Proof of Proposition 2. Write $v = s S \Pi w$. The sign s is irrelevant to the quadratic form ($s^2 = 1$), and the permutation Π is independent of w , so we may compute the law of $y = S \Pi w$: since $\Pi w \stackrel{d}{=} \sqrt{d} u$ for u uniform on the unit sphere of $\mathbb{R}^d$ (Haar composed with a uniform permutation is Haar), y is the vector of the first m coordinates of $\sqrt{d} u$. Decompose $u = (u_m, u_\perp)$ with

$\beta = \|u_m\|^2 \sim \text{Beta}(\frac{m}{2}, \frac{d-m}{2})$ and $\hat{u} = u_m/\sqrt{\beta}$ uniform on S^{m-1} , independent of β (a classical property of the sliced sphere). Then

$$v^\top Av = d\beta (\hat{u}^\top A\hat{u}).$$

For $\hat{u}$ uniform on S^{m-1} , invariant theory gives $\mathbb{E}[\hat{u}_j\hat{u}_l\hat{u}_p\hat{u}_q] = \frac{\delta_{jl}\delta_{pq} + \delta_{jp}\delta_{lq} + \delta_{jq}\delta_{lp}}{m(m+2)}$, so $\mathbb{E}[\hat{u}^\top A\hat{u}] = \text{tr}(A)/m$ and $\mathbb{E}[(\hat{u}^\top A\hat{u})^2] = \frac{\text{tr}(A)^2 + 2\text{tr}(A^2)}{m(m+2)}$. The beta moments are $\mathbb{E}[\beta] = m/d$ and $\mathbb{E}[\beta^2] = \frac{m(m+2)}{d(d+2)}$. Hence

$$\begin{aligned}\mathbb{E}[v^\top Av] &= d \cdot \frac{m}{d} \cdot \frac{\text{tr}(A)}{m} = \text{tr}(A), \\ \mathbb{E}[(v^\top Av)^2] &= d^2 \frac{m(m+2)}{d(d+2)} \cdot \frac{\text{tr}(A)^2 + 2\text{tr}(A^2)}{m(m+2)} \\ &= \frac{d(\text{tr}(A)^2 + 2\text{tr}(A^2))}{d+2},\end{aligned}$$

and therefore

$$\text{Var}[v^\top Av] = \frac{d(\text{tr}(A)^2 + 2\text{tr}(A^2))}{d+2} - \text{tr}(A)^2 = \frac{2}{d+2} (d\text{tr}(A^2) - \text{tr}(A)^2).$$

For $g \sim \mathcal{N}(0, I_m)$ the Gaussian fourth moments $\mathbb{E}[g_j g_l g_p g_q] = \delta_{jl}\delta_{pq} + \delta_{jp}\delta_{lq} + \delta_{jq}\delta_{lp}$ give $\text{Var}[g^\top Ag] = 2\text{tr}(A^2)$. The ratio is $\frac{d\text{tr}(A^2) - \text{tr}(A)^2}{(d+2)\text{tr}(A^2)} = \frac{d - \kappa(A)}{d+2}$ with $\kappa(A) = \text{tr}(A)^2/\text{tr}(A^2) \geq 1$ for $A \succeq 0$, $A \neq 0$ (Cauchy-Schwarz on the eigenvalues) and $\kappa(A) \leq \text{rank}(A) \leq m$. $\square$

Lemma (covariances without per-molecule permutations). The following classical computation explains what the per-molecule permutations buy. Consider the permutation-free variant in which two molecules use the shared direction w (or distinct orthogonal directions $w \perp w'$) directly, without Π_i ; define $\Delta(A_i, A_j) = d\text{tr}(A_i A_j) - \text{tr}(A_i)\text{tr}(A_j)$. Then sharing one direction gives

$$\text{Cov}(w^\top A_i w, w^\top A_j w) = \frac{2}{d+2} \Delta(A_i, A_j),$$

while distinct orthogonal directions give

$$\text{Cov}(w^\top A_i w, w'^\top A_j w') = -\frac{1}{d-1} \cdot \frac{2}{d+2} \Delta(A_i, A_j).$$

$$\mathbb{E}[w^\top A_i w w^\top A_j w] = d^2 \frac{\text{tr}(A_i)\text{tr}(A_j) + 2\text{tr}(A_i A_j)}{d(d+2)} = \frac{d(\text{tr}(A_i)\text{tr}(A_j) + 2\text{tr}(A_i A_j))}{d+2},$$

so

$$\begin{aligned}\text{Cov} &= \frac{d\text{tr}(A_i)\text{tr}(A_j) + 2d\text{tr}(A_i A_j) - (d+2)\text{tr}(A_i)\text{tr}(A_j)}{d+2} \\ &= \frac{2}{d+2} (d\text{tr}(A_i A_j) - \text{tr}(A_i)\text{tr}(A_j)).\end{aligned}$$

Proof of Lemma A, orthogonal directions. Let (u, u') be the first two columns of a Haar-distributed orthogonal matrix. Conditional on u , the vector u' is uniform on the unit sphere of $u^\perp$ (dimension $d-1$), so with $P_u = I - uu^\top$,

$$\mathbb{E}[u'^\top A_j u' | u] = \frac{\text{tr}(P_u A_j P_u)}{d-1} = \frac{\text{tr}(A_j) - u^\top A_j u}{d-1}.$$

Therefore, using $\mathbb{E}[u^\top A_i u] = \text{tr}(A_i)/d$ and $\mathbb{E}[(u^\top A_i u)(u^\top A_j u)] = \frac{\text{tr}(A_i)\text{tr}(A_j) + 2\text{tr}(A_i A_j)}{d(d+2)}$,

$$\begin{aligned}\mathbb{E}[u^\top A_i u u'^\top A_j u'] &= \frac{\text{tr}(A_j) \mathbb{E}[u^\top A_i u] - \mathbb{E}[(u^\top A_i u)(u^\top A_j u)]}{d-1} \\ &= \frac{1}{d-1} \left[\frac{\text{tr}(A_i)\text{tr}(A_j)}{d} - \frac{\text{tr}(A_i)\text{tr}(A_j) + 2\text{tr}(A_i A_j)}{d(d+2)} \right].\end{aligned}$$

Subtracting the product of the means $\frac{\text{tr}(A_i)}{d} \frac{\text{tr}(A_j)}{d}$ and simplifying,

$$\text{Cov}[u^\top A_i u, u'^\top A_j u'] = -\frac{2}{d(d-1)(d+2)} \left(\text{tr}(A_i A_j) - \frac{1}{d} \text{tr}(A_i) \text{tr}(A_j) \right).$$

Scaling both probes by $\sqrt{d}$ multiplies each quadratic form by d and the covariance by d^2 :

$$\begin{aligned} \text{Cov}[w^\top A_i w, w'^\top A_j w'] &= -\frac{2}{(d-1)(d+2)} \left(d \text{tr}(A_i A_j) - \text{tr}(A_i) \text{tr}(A_j) \right) \\ &= -\frac{1}{d-1} \cdot \frac{2}{d+2} \Delta(A_i, A_j). \end{aligned}$$

As a check, setting $A_i = A_j = A$ and summing over a full orthonormal basis $\{w_1, \dots, w_d\}$ gives $\text{Var}[\sum_k w_k^\top A w_k] = d \cdot \frac{2\Delta}{d+2} + d(d-1) \cdot \left(-\frac{2\Delta}{(d-1)(d+2)} \right) = 0$: a complete basis recovers the trace exactly, and the lemma’s constants are the finite- P residue of that exactness. $\square$

Proof of the variance ratio (5). By Proposition 3, all cross-molecule covariances vanish under the OPS law, so the decomposition (4) retains only the diagonal terms, each given by Proposition 2 (with $m = 3N_i$ and the molecule’s own A_i ; the padded coordinates contribute nothing). Dividing by the PHL variance $\frac{1}{B^2} \sum_i 2\text{tr}(A_i^2)$ and collecting terms yields (5). For the shared Gaussian pool, adding the norm-channel covariances of Proposition 3 over the N_{same} same-direction pairs gives the penalty stated in Section 3.3. $\square$

B Empirical unbiasedness check

At model initialization (Kaiming-normal weights, the training-time distribution) we draw 300 independent minibatch HVP-loss estimates per probe law on each of three training minibatches and compare means; the analytic proof (Proposition 1) is the source of truth, and this check validates the implementation (seeded, logged in `logs/unbiasedness_check.json`). Per-batch means (hartree²; standard errors of the mean in parentheses):

Law	batch 1	batch 2	batch 3
PHL iid Gaussian	0.1645 (0.0003)	0.1660 (0.0003)	0.1660 (0.0003)
OPS $P=16$	0.1641 (0.0003)	0.1673 (0.0003)	0.1659 (0.0003)
OPS $P=32$	0.1646 (0.0003)	0.1667 (0.0003)	0.1656 (0.0003)
OPS $P=48$	0.1647 (0.0003)	0.1672 (0.0003)	0.1662 (0.0003)
Rademacher	0.1642 (0.0002)	0.1665 (0.0002)	0.1663 (0.0002)

All laws agree within ~ 2 standard errors of each other on every batch; across the 15 comparisons the largest deviation is 2.7 standard errors, consistent with the expected maximum of 15 Gaussian-distributed differences. The estimator standard deviations are also informative: the OPS laws have loss-estimator standard deviation 0.0046 versus 0.0048 for PHL (ratio 0.92), and Rademacher 0.0040 (ratio 0.84), matching the theoretical per-direction analysis of Proposition 2 plus the small sharing penalty of Lemma A.

C Split composition and provenance

The authors’ data pipeline loads every `.h5` file in the data directory (six interleaved files = three source datasets, each present as a byte-identical duplicate pair), pools all conformers, shuffles with the run seed, and splits 80/10/10 at the conformer level. We reproduce this pipeline verbatim and recover each held-out conformer’s source dataset by MD5 hashing of its coordinate array against the three unique source files. For seed 0 the held-out evaluation split contains 26,373 molecules: 6,982 RTP (Test), 6,926 IRC, and 12,465 NMS; the training split contains 210,979 molecules (55,689 RTP, 55,217 IRC, 100,073 NMS). Other seeds give equivalent compositions. Because the split is conformer-level on the pooled (duplicated) files, a molecule’s two file copies can in principle land in different splits; this behavior is inherited unchanged from the reference implementation and is identical for both methods, so paired comparisons are unaffected. We tag every evaluation molecule by its source dataset and report all accuracy metrics per (source $\times$ property) cell.

D Additional experimental details

Hardware. NVIDIA H200 GPUs (one process per run, two runs share a GPU for two of the ten main runs; GPU sharing affects only wall-clock measurements, which we report from dedicated-GPU runs), float64 throughout, PyTorch 2.13 + TorchANI 2.2.4 with the authors’ modified `torchani/utils.py` and `torchani/data/__init__.py` (byte-identical to the workspace install; see `codebase_notes.md`).

Probe accounting. The training loop’s per-epoch probe count (Figure 2, right) is the number of distinct probe-vector HVP evaluations consumed by parameter updates: B per step for PHL (one fresh Gaussian probe per molecule; 210,979 per epoch = 528 steps at $B=400$ modulo the final partial batch), $P=32$ per step for OPS. Audit probes (used only to measure variance at frozen checkpoints, never to update parameters) are drawn from a separate generator and excluded from the metric.

Variance audit details. At each audit epoch we freeze the model and evaluate, for each of 8 fixed training minibatches and 3 pure-source minibatches (RTP/IRC/NMS), the gradient of the 0.09-weighted HVP loss w.r.t. all 97,924 trainable parameters: 48 draws under the audited law, 48 under the PHL law, and 256 additional PHL-law draws averaged into the oracle $\bar{g}$. The reported variance is $\sum_p \text{Var}[g_p - \bar{g}_p]$ over parameters, per law, and the ratio is (audited law)/PHL. The oracle’s residual error inflates both measured variances equally at leading order and biases the ratio toward 1 by $\sim 1/256$, negligible at our precision.

Reproducibility. Data splits, model initialization, and probe draws are all governed by the run seed through Python/NumPy/PyTorch seeded generators; both methods at a given seed see byte-identical splits and initializations. Probe streams use dedicated `torch.Generators` and are logged implicitly through the seed chain.

Schedules. Main comparison: 320 epochs, 5 seeds, both methods. Ablations: 160 epochs, 2 seeds (gauss-pool, no-share, Rademacher). E–F control: 320 epochs, 2 seeds. All other hyperparameters identical to the authors’ notebook (batch 400, AdamW/SGD with ReduceLROnPlateau factor 0.5 patience 100, $\lambda_F = 0.30$, $\lambda_H = 0.09$, early stop at $\text{LR} < 10^{-5}$, never triggered at these schedules).